6.1 Exponents

Lesson Introduction

Benchmark	MA.6.NSO.3.3 Evaluate positive rational numbers and integers with natural number exponents.	Learning Targets	• I understand the role of a base and an exponent in the exponential form of a numeric expression. • I can write a numeric expression using repeated multiplication in exponential form. • I can write an expression in exponential form as a numeric expression using repeated multiplication. • I can evaluate a positive rational number raised to a power.

	Building On	Working Toward
Benchmark Coherence	N/A	MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

Essential Questions	• How do you write the exponential form of a numeric expression using repeated multiplication? • How do you write repeated multiplication as the exponential form of a numeric expression? • How do you evaluate positive rational numbers raised to powers?
Lesson Narrative	Students learn how to rewrite exponential expressions as repeated multiplication. They learn that the exponent indicates how many times to multiply the base by itself. Students also rewrite repeated multiplication of the same positive rational numbers as numeric exponential expressions. Finally, students evaluate positive rational numbers raised to exponents.

Component Summary				
	6.1.1 Warm-Up (~5 min.)	Relate graphic designing to studying mathematics *(SE p. 233)*	6.1.4 Collaboration (10 min.)	Collaboratively evaluate exponential expressions and recognize common errors *(SE p. 235)*
	6.1.2 Guided Instruction (10 min.)	Represent repeated multiplication as an exponential expression and evaluate *(SE p. 234)*	6.1.5 Try It (5 min.)	Practice evaluating exponential expressions *(SE p. 236)*
	6.1.3 Guided Practice (5 min.)	Practice writing exponential expressions using repeated multiplication and vice versa *(SE p. 235)*	6.1.6 Your Turn! (5-10 min.)	Identify and write equivalent expressions *(SE p. 237)*
	6.1.7 Wrap-Up (~5 min.)	Error analysis of student work evaluating exponential expressions *(SE p. 237)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Base • Exponent • Evaluate Additional Terms: N/A	• (Optional) Large chart paper for anchor chart	• 6.1.1 Warm-Up contains a video to accompany the question prompts.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>• Students may use terms like “solve” when referring to evaluating expressions.
• Students may confuse the directions “evaluate” and “rewrite” (in an equivalent form).
• Students may multiply the base by the exponent when asked to evaluate an exponential expression or rewrite an equivalent expression.</td><td>• Students are expected to multiply positive rational numbers with procedural fluency by the end of Grade 6 (MA.6.NSO.2.2). Consider ways to support students in developing this fluency with multiplying positive whole numbers, decimals, and fractions when addressing this benchmark instead of giving them a calculator.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
In 6.1.6 Your Turn!, ask students what they notice and wonder about the different forms of the equivalent expressions. These discussions will help students make connections and build their understanding.</td><td>MA.K12.MTR 4.1: Discussion and Reflection
6.1.4 Collaboration includes error analysis problems. In question 2, students determine if the provided student work is correct, and if it is incorrect, they describe and correct the error(s). In question 3, they determine if the expressions are equivalent and evaluate each expression to justify their claim.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider creating an anchor chart to help students learn key terms in 6.1.2 Guided Instruction. Include the definitions and examples to provide a visual entry point.

Use the Think Aloud strategy to debrief 6.1.3 Guided Practice to provide an auditory entry point.

Consider allowing students who struggle writing explanations to use an audio recorder in the 6.1.7 Wrap-Up.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

6.1.1 Warm-Up: Graphic Designer

Component Overview: *Play the career video featuring a graphic designer for the class. Then, the students will answer three questions relating the messages from the video to their own life experiences and goals.*

6.1.1 Warm-Up: Graphic Designer

1. Maurice Woods is a graphic designer and founder of the Inneract Project. He says, "Design is really about a process of thinking creatively and critically about how to solve something in a way that not only makes sense to people, but produces a solution to a problem." How does that process relate to your study of mathematics?
   Answers will vary. In math class I use my understanding of both math and situations I have experienced to come up with a logical solution to the problem.

2. Maurice states that one of the skills he brought with him from his basketball career to his design career was his experience with teamwork and what it means to be flexible and compromise. Describe an experience when teamwork was needed for you to reach a goal.
   Answers will vary. As a soccer player I could run down the field and try to score a goal on my own. If I work with my teammates to pass the ball moving down the field, that increases the chances of making a goal.

3. Maurice said, "I want to be remembered for giving back to the community and giving back to young people ... I get more joy out of that than any project I work on." What do you want to be remembered for in life?
   Answers will vary. I want to be remembered for helping others in need. I want to be remembered for inventing something that helps people live healthier lives.

Consider an extension project in which students explore career(s) they are interested in pursuing and research how their desired careers use mathematical content and/or practices.

Prompt students to share their responses with a partner or monitor for strong responses and have students share whole group.

6.1.2 Guided Instruction: Expressions with Exponents

Component Overview: *Students represent repeated multiplication as exponential expressions and evaluate exponential expressions. Students develop an understanding of the key terms "base," "exponent," and "evaluate."*

6.1.2 Guided Instruction: Expressions with Exponents

1. The principal at Fitzgerald Middle School has been tasked with contacting everyone at the school regarding an upcoming change. The principal uses a phone tree to contact the staff. The principal calls 2 people, then those 2 people will call 2 other people. This process repeats until everyone has been contacted.

   a. Fill in the blanks for the phone tree.

Phone Tree	
First round of calls:	2
Second round of calls:	$2 \cdot 2$
Third round of calls:	$2 \cdot 2 \cdot 2$
Fourth round of calls:	$2 \cdot$ _2__ $\cdot$ _2__ $\cdot$ _2__
Fifth round of calls:	$2 \cdot$ _2__ $\cdot$ _2__ $\cdot$ _2__ $\cdot$ _2

   b. What do you notice about the phone tree?
   Answers will vary. Each round increases by a multiple of two.

   c. How many people were contacted on the fifth round?
   32

The fifth round of calls can be rewritten in exponential form as 2^5.

   d. How does the exponential expression for the fifth round of calls relate to the repeated multiplication written in the phone tree?
   In the exponential expression, the 5 indicates the number of times that 2 is multiplied in the phone tree.

Remind students to use talk moves, such as revoicing and thinking aloud, when working with their partners and during the whole-class debrief discussion.

How would the expression using repeated multiplication change if the base was 5 and the exponent was 2, as in 5^2?

6.1.2 Guided Instruction: Expressions with Exponents (continued)

The numeric expression $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$ can be rewritten in exponential form as 2^5, where 2 is the ***base*** and 5 is the ***exponent***.

The ***base*** of an exponent is the number used as a factor in exponential form.

The ***exponent*** is the number of times the base occurs as a factor.

Exponents are also commonly referred to as powers. For example, 2^5 is read as "2 to the 5^{th} power."

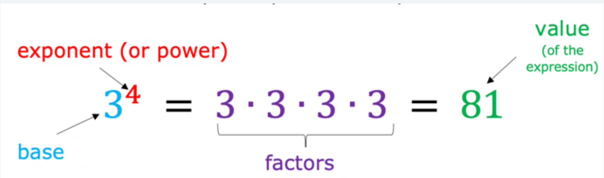
2. Complete the statements using the definitions of base and exponent.
   a. In the exponential expression 9^3, the 9 is the
      - ○ base
      - ○ exponent

      of the expression and the 3 is the
      - ○ base.
      - ○ exponent.
   b. The exponential expression 9^3 has the same value as
      - ○ $3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$.
      - ○ $9(3)$.
      - ○ $(9)(9)(9)$.

In mathematics, **evaluate** is an instruction that means "find the value of."

3. Evaluate 9^3.
   729

Consider creating an anchor chart to help students learn key terms. Include the definitions and examples to provide a visual entry point. Also, consider using different colors to support students in differentiating between the base and the exponent. Pay attention to precision in how students are writing the exponent as a superscript. This visual representation can also be found later in this unit.



Consider asking students the following:

- *How have you used the word "evaluate" in the past? How is this different from the definition we use in math?*
- *In your own words, if the question asks you to evaluate, what would you do?*

6.1.3 Guided Practice: Equivalent Expressions

Component Overview: *Students practice rewriting exponential expressions using repeated multiplication and vice versa. Students can complete this activity independently and peer review their responses with their partner.*

6.1.3 Guided Practice: Equivalent Expressions

1. Write each numeric expression in exponential form.

   a. $5 \cdot 5 \cdot 5 \cdot 5$

   5^4

   b. $\left(\frac{6}{7}\right) \times \left(\frac{6}{7}\right) \times \left(\frac{6}{7}\right)$

   $\left(\frac{6}{7}\right)^3$

   c. $(3.25)(3.25)$

   $(3.25)^2$

2. Rewrite each expression given in exponential form as repeated multiplication.

   a. 7^5

   $7 \cdot 7 \cdot 7 \cdot 7 \cdot 7$

   b. $(1.5)^2$

   $(1.5)\,(1.5)$

   c. $\left(\frac{3}{4}\right)^3$

   $\left(\frac{3}{4}\right) \cdot \left(\frac{3}{4}\right) \cdot \left(\frac{3}{4}\right)$

Consider asking students the following:

- *When rewriting an expression with the same number being multiplied by itself repeatedly, how do you know what exponent to use?*
- *When expanding an expression given in exponential form, how do you know what the factors are and how many factors to include?*

Challenge learners to evaluate the expressions and write an explanation of the reasoning behind their answers if they finish early.
This will be addressed in the next components.

6.1.4 Collaboration: Evaluating Exponential Expressions

Component Overview: *In this activity, students will transition from rewriting to evaluating expressions. This component includes error analysis problems to highlight and address common misconceptions. Students will also explore equivalent expressions written in different forms.*

6.1.4 Collaboration: Evaluating Exponential Expressions

1. With your partner, discuss the four expressions in the table.

Expression A	$(2)(2)(2)(2)$
Expression B	(2^4)
Expression C	16
Expression D	2×4

   a. Which expression does not represent the same value as the others?
   Expression D, 2×4

   b. Explain your answer or show the work that led to your conclusion.

Expression A	Expression B	Expression C	Expression D
$(2)(2)(2)(2)$ $= (4)(2)(2)$ $= (8)(2)$ $= 16$	(2^4) $= (2)(2)(2)(2)$ $= (4)(2)(2)$ $= (8)(2)$ $= 16$	*The expression* 16	2×4 $= 8$

   c. Explain what mathematics process should be done if the directions state to **evaluate** an expression.
   Answers will vary. Evaluate an expression means "find the value of" the expression. Use the order of operations to find the value of the expression.

2. Marta's work for evaluating the expression 3^2 is shown:

$$3^2 = 3 \times 2 = 6$$

   a. Discuss Marta's work with your partner.
   Partner discussion.

Have students practice interpreting these mathematical statements, particularly reading the same statement in multiple ways.

Prompt students to discuss the difference between a^b and $a \times b$, or have them provide examples to prove their explanation.

b. Explain whether her work is correct or incorrect. If it is correct, describe how you know. If it is incorrect, describe her error and provide the correct evaluation of the expression.
Answers will vary. Martha is incorrect. She multiplied the base by the power. The 2 in the exponential expression is the power and the 3 is the base. This means that 2 factors of 3 are to be multiplied. The correct answer is: $3 \cdot 3 = 9$.

Ask student how they might explain to Marta the correct way to complete the math problem.

3. Determine if each equation in the table is true or false. Show your work to support your answer.

Equation	True or False?	Work
$16^1 = 8^2$	*False*	$16^1 = 16$ $8^2 = 8 \cdot 8$ $= 64$ $16 \neq 64$
$3+3+3 = 3^3$	*False*	$3+3+3=9$ $3^3 = 3 \cdot 3 \cdot 3$ $= 9 \cdot 3$ $= 27$ $9 \neq 27$
$\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2} = 4 \cdot \frac{1}{2}$	*False*	$\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}\cdot\frac{1}{2}\cdot\frac{1}{2}$ $= \frac{1}{4}\cdot\frac{1}{4} = \frac{1}{16}$ $4 \cdot \frac{1}{2} = \frac{4}{2} = 2$ $\frac{1}{16} \neq 2$
$(5)^3 = 5 \cdot 5 \cdot 5$	*True*	$(5)^3 = 5 \cdot 5 \cdot 5$ $= 25 \cdot 5$ $= 125$

6.1.5 Try It: Evaluating Exponential Expressions

Component Overview: *Students practice evaluating exponential expressions. The bases represent a variety of rational numbers (a fraction, whole number, and decimal).*

6.1.5 Try It: Evaluating Exponential Expressions

1. Evaluate each expression written in exponential form.

   a. $\left(\frac{2}{5}\right)^2$

   $\frac{4}{25}$

   b. 4^5

   1,024

   c. $(6.2)^3$

   238.328

Students who struggle may benefit from writing each expression out using repeated multiplication before evaluating the problem.

6.1.6 Your Turn!

Component Overview: *Students will practice identifying equivalent expressions, rewriting expressions, and evaluating them. Circulate while students work to monitor progress, assist struggling students, and determine which students you will call on to share during the whole-class debrief.*

6.1.6 Your Turn!

1. For each set of expressions, circle the equivalent expressions.

   a. $(4)^3$ $\quad(3)^4\quad$ $(4)(4)(4)$ $\quad 4+4+4$

   b. $\left(\frac{5}{3}\right)^2$ $\quad\left(\frac{3}{5}\right)^2\quad$ $\frac{10}{6}$ $\quad\frac{25}{9}$

Ask students what they notice and wonder about the different forms of the equivalent expressions. These discussions will help students make connections and build their understanding.

2. Complete the table by rewriting the numeric expression in the form indicated. Then, find the value of each expression.

Exponential Form	Expanded Form	Value
11^3	$(11)(11)(11)$	$1{,}331$
$\left(\frac{7}{15}\right)^3$	$\left(\frac{7}{15}\right)\cdot\left(\frac{7}{15}\right)\cdot\left(\frac{7}{15}\right)$	$\frac{343}{3{,}375}$
$(5.2)^2$	$(5.2)\cdot(5.2)$	27.04
$\left(\frac{4}{5}\right)^4$	$\frac{4}{5}\cdot\frac{4}{5}\cdot\frac{4}{5}\cdot\frac{4}{5}$	$\frac{256}{625}$

6.1.7 Wrap-Up: Ticket Out the Door

Component Overview: *Students demonstrate their understanding of the lesson learning targets by analyzing samples of student work for errors. Consider using data from this formative assessment to determine which students need support before the next lesson.*

6.1.7 Wrap-Up: Ticket Out the Door

Jose and Georgette were asked to each roll a number cube five times and record the number from each roll onto a chart. The number that Jose rolls represents the base of an exponential expression, and Georgette's roll represents the exponent of the expression. Their table below shows the results of rolling the number cubes five times.

Roll	Jose's Number Cube	Georgette's Number Cube	Exponential Expression
1	5	3	5^3
2	5	4	5^4
3	6	2	6^2
4	4	5	4^5
5	1	4	1^4

Jose conjectures that the value of the exponential expression from Roll 3 is the greatest because the exponential expression that has the largest base is always greater.	Georgette conjectures that the value of the exponential expression from Roll 4 is the greatest because the exponential expression that has the largest exponent is always greater.

1. Determine if either Jose or Georgette were right. If neither are correct, then which exponential expression has the largest value?
   In this case, Georgette is correct because $4^5 = 1{,}024$, which is greater than $5^4 = 625$.

2. Determine if it is possible that Jose's conjecture or Georgette's conjecture is always true using the exponential expressions 2^5 and 5^2. Explain your findings.
   Answers will vary. $2^5 = 32$ and $5^2 = 25$. Based on the answers, Jose's conjecture is not always true because although 5 is the larger base, the value of the exponential expression 5^2 is less. Georgette's conjecture is also not always true. For example, if the base is 1, the overall value will always be 1, even if raised to a higher power.

Provide students with scaffolding prompts if they are unsure where to begin.

- *Read the information for context.*
- *After reading, what is the question asking?*
- *Compare what you think the answer is to Jose's and Georgette's responses. Then answer questions 1 and 2 based on your findings.*